21-242 Recitation 12

Fundamental theorems — Q3. T A surjective implies A has a right inverse

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that T_A surjective implies A has a right inverse.

Course-level solution

Let $T_A : \mathbb{F}^n \to \mathbb{F}^m$ be surjective. For each standard basis vector $e_i \in \mathbb{F}^m$ choose a preimage $\vec{v}_i \in \mathbb{F}^n$ with $A\vec{v}_i = e_i$. Let C be the $n \times m$ matrix with columns $\vec{v}_1, \ldots, \vec{v}_m$. Then

$$ACe_i = A\vec{v}_i = e_i \qquad (i = 1, \ldots, m),$$

so $AC = I_m$. (The map T_C is a right inverse of T_A.)

Later-course solution (using ideas from the rest of the handout)

Surjectivity means $\operatorname{rank}(A) = m$, so $m \leq n$ and $\operatorname{RREF}(A) = \begin{pmatrix} I_m & F \end{pmatrix}$ after row operations: $GA = \begin{pmatrix} I_m & F \end{pmatrix}$. A particular right inverse is $C = \begin{pmatrix} I_m \\ 0 \end{pmatrix} G$ when we pad with free-variable zeros after undoing G, or more invariantly: choose any complementary subspace to $\ker T_A$ (Dimension Q6); T_A restricted there is an isomorphism $\mathbb{F}^n$/direction of $\ker \to \mathbb{F}^m$, and the inverse of that isomorphism, extended by 0 on a complement, is a right inverse. The preimage-of-basis argument above is the one to write at this point in the recitation.